

PROBLEM SET 1: THE RAMSEY MODEL

PROBLEM I – THE OPEN-ECONOMY RAMSEY MODEL WITH INSTALLATION COSTS FOR CAPITAL

We consider an economy which produces a single homogeneous good and can borrow and lend at a fixed rate r (in terms of this good which we will take as the numéraire) from the rest of the world. Its total labor force is normalized to 1 so that its production function is

$$Y_t = AK_t^\alpha.$$

The representative consumer's utility function is

$$V = \int_0^{+\infty} e^{-\rho t} C_t^\beta dt. \quad (1)$$

We assume that the subjective discount rate is equal to the world interest rate:

$$r = \rho$$

The equation for accumulating capital is

$$\dot{K}_t = I_t, \quad (2)$$

where I_t = investment. To install I units of capital we need to spend $f(I)$ units of the final good, in addition to the I units that are needed for investment. We assume $f(0) = f'(0) = 0$, $f'(x) > 0$ for $x > 0$, $f'(x) < 0$ for $x < 0$, $f''(x) > 0$, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow +\infty} f'(x) = +\infty$,

- 1 – What is the surplus of the trade balance at an instant t ?
- 2 – Let W_t be the net stock of foreign assets owned by this economy (in terms of the numéraire). What is the evolution of W over time?
- 3 – Show that if we impose the solvency condition

$$\lim e^{-rt} W_t > 0,$$

then the country has an intertemporal budget constraint given by

$$W_0 + \int_0^{+\infty} e^{-rt} (Y_t - C_t - I_t - f(I_t)) dt \geq 0.$$

- 4 – Show that the optimal investment strategy must maximize the following criterion:

$$\tilde{W} = \max \int_0^{+\infty} (Y_t - I_t - f(I_t)) e^{-rt} dt$$

subject to (2)

- 5 – Show that the optimal investment strategy is given by

$$I_t = g(q_t)$$

where $g()$ is a strictly increasing function such that $g(1) = 0$ and q_t satisfies

$$-\dot{q}_t + r q_t = \alpha A K_t^{\alpha-1}$$

- 6 – What is the economic interpretation of q ? Show that it is equal to the present discounted value of future marginal products of capital net of depreciation.
- 7 – What is the level of capital in steady state? Does the corresponding condition sound familiar?
- 8 – Draw then phase diagram of this problem. How would you eliminate the trajectories that do not converge to the steady state?

9 – Using this diagram, describe how investment reacts to a permanent increase in A at date $t = T$, when this increase is anticipated at date $t = 0$. How would you interpret this economically?

10 – Show that the optimal consumption behavior maximizes (1) subject to

$$\int_0^{+\infty} C_t e^{-rt} dt \leq W_0 + W^*,$$

where W^* is the value of $\tilde{W}$ at the optimum investment strategy.

11 – What is the solution to this problem?

12 – Assuming that at $t = 0$ the economy is in steady state with $W_0 = 0$, what is the reaction of the trade balance to the anticipated increase in A discussed in question 9?

PROBLEM II – THE GOLDEN RULE

We consider an economy with the production function

$$Y_t = F(K_t, N_t),$$

where N_t = employment = population and $dN/dt = nN_t$.

The capital accumulation equation is

$$\dot{K}_t = Y_t - C_t - \delta K_t.$$

1 – Rewrite these equations in per-capita term.

2 – Characterize a balanced growth path.

3 – What is the ratio K/N which maximizes consumption per capita? This is called the "Golden rule" level.

4 – Compare the net marginal product of capital with the population growth rate at the golden rule.

5 – Assume factors are paid their marginal product. Compare the share of profits in total income to the share of investment in output at the golden rule.

6 – Assume 'society' maximizes

$$V = \int_0^{+\infty} e^{-\rho t} N_t^\alpha \frac{\left(\frac{C_t}{N_t}\right)^\beta - 1}{\beta} dt,$$

How would you interpret such a utility function? Show that for the problem to be meaningful we need $\alpha n < \rho$. What happens if this is not the case?

7 – Compare the long-run K/N level to the Golden rule in this case.

8 – How does α affect the long-term stock of capital? Explain.

PROBLEM III – DECENTRALIZATION OF THE EQUILIBRIUM

We consider the standard Ramsey model, and to simplify we assume no technical progress and no population growth. Thus we have

$$Y_t = F(K_t, L_t), \tag{3}$$

and in equilibrium $L_t = \bar{L} = 1$.

We assume that at any date t there is a continuum of representative firms of mass 1 which all have the same production function and the same capital and employment as in the aggregate economy.

$$\dot{K}_t = Y_t - C_t - \delta K_t.$$

Finally there is a continuum of identical consumers with mass 1, each endowed with L units of labor and their utility function is

$$U = \int_0^{+\infty} u(C_t) e^{-\rho t} dt.$$

At each date there is a competitive market for labor and the wage is w_t . Furthermore, at each date people can borrow and lend at a competitive rate r_t . The supply of loanable funds must be equal to the demand for loanable funds. Finally the firms are owned by the consumers.

- 1 – Show that the consumers' disposable income is equal to Y_t .
- 2 – Show that the value of a firm V_t evolves according to

$$r_t V_t = F(K_t, L_t) - w_t L_t - I_t + \dot{V}_t, \quad (4)$$

where I_t are the investment expenditures of the firm at date t . From the point of view of the firm one has $\dot{K}_t = I_t - \delta K_t$, and furthermore it treats L_t not as fixed but as a choice variable.

- 3 – What is the fundamental value of V_t at t ?
- 4 – Assume that V_t is always equal to its fundamental value and that firms set their optimal investment strategy so as to maximize V_t . Show that at each date t one must have

$$\frac{\partial F}{\partial K}(K_t, L) = r_t + \delta.$$

Interpret.

- 5 – What is the wage level at t given K_t ?
- 6 – Let W_t be the net stock of assets owned by the representative consumers (that is, the money owed by firms to the consumers at t , expressed in terms of the output good). What is the evolution of W_t over time?
- 7 – Assume that we impose the solvency condition

$$\lim e^{-rt} W_t > 0,$$

what is the consumer's intertemporal budget constraint?

- 8 – Solve the consumer's optimization problem and derive the Euler equation.
- 9 – Show that in equilibrium the Euler equation is the same as on the optimal trajectory and therefore that the Ramsey optimum is also a competitive equilibrium.
- 10 – Show that $V_t = K_t$. Explain. (Hint: manipulate (4) using the equilibrium factor prices and the fact that $F(\cdot)$ is homogeneous of degree one).